

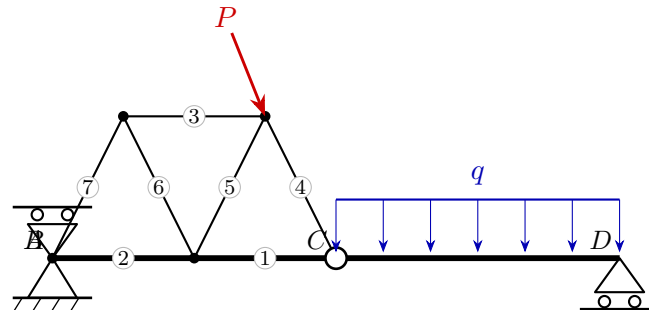
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam Group A

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Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k , reactions a , interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \Rightarrow \text{statically determinate}$$

2. Support reactions and hinge forces

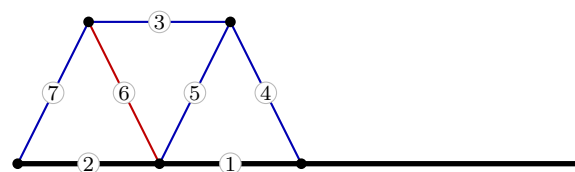
$$A: R_x = -8, R_y = 12.5 \quad D: R_x = 0, R_y = 18.5 \quad B: R_x = 0, R_y = 1$$

$$C: H_x = 8, H_y = -19$$

3. Truss member forces

Member	Force S [kN]	Type
1	1.5	tension
2	0.5	tension
3	-1	compression
4	-21.24	compression
5	-1.12	compression
6	1.12	tension
7	-1.12	compression

red: tension (+) blue: compression (-)



4. Section-force diagrams of the beams

